Object-Oriented Programming Case Study

PROBLEM STATEMENT: Smart Parking Lot Management System

Technical Analysis & Implementation Report

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1. Project Overview

The **Smart Parking Lot Management System** is a console-based application designed to simulate real-world parking management. It demonstrates core Object-Oriented Programming (OOP) concepts through a structured and interactive system.

The system manages parking allocation, tracks vehicle entry and exit, and calculates parking fees dynamically based on the type of parking spot used.

Key Features:

- **Total Capacity:** 20 parking spots
  - 12 Standard spots
  - 4 Disabled spots
  - 4 VIP spots
- **Real-time tracking** of occupancy status
- **Dynamic pricing system** based on spot type
- **Interactive menu-driven interface**

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2. System Design & Class Structure

The system is built using a hierarchical design based on OOP principles:

Base Structure:

- A central abstract concept represents a parking spot.
- It defines common attributes like:
  - Spot ID
  - Occupancy status

Specialized Types:

Three types of parking spots extend the base structure:

- **Standard Spot** – Regular parking
- **Disabled Spot** – Discounted parking for accessibility
- **VIP Spot** – Premium parking with higher pricing

Each type has its own pricing logic and behavior, enabling flexible and scalable design.

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3. Core Concepts Demonstrated

Polymorphism

The system treats all parking spots uniformly while allowing each type to behave differently when needed (e.g., fee calculation). This ensures flexibility without complex conditional logic.

Dynamic Memory Management

Parking spots are created dynamically during runtime, allowing efficient use of memory and scalability. Proper cleanup ensures no memory leaks.

Use of Standard Library (STL)

- A dynamic container is used to store all parking spots
- Efficient iteration and search operations are implemented
- Ensures clean and maintainable code structure

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4. System Architecture

The application follows a layered structure:

- **Main Program**
  - Controls user interaction through a menu
- **Parking Lot Manager**
  - Handles allocation, tracking, and operations
- **Parking Spot Types**
  - Represent different categories (Standard, Disabled, VIP)

Workflow:

1. System initializes all parking spots
2. User selects actions (park, exit, view status)
3. System allocates the first available spot of the requested type
4. On exit, the system calculates the fee based on duration and type
5. Status updates are reflected in real time

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5. Functional Behavior

Typical Usage Flow:

1. System starts and initializes all parking spots
2. User parks a vehicle by selecting a spot type
3. The system assigns an available spot
4. User can view current parking status at any time
5. When exiting, the system:
   - Calculates parking duration
   - Applies pricing rules
   - Displays total fee
6. Spot becomes available again

Pricing Model:

- **Standard:** Hourly rate
- **Disabled:** Reduced hourly rate
- **VIP:** Base charge + additional hourly rate

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6. Performance & Complexity

- **Search Operations:** Linear time complexity (O(n))
- **Status Updates:** Constant time (O(1))
- **Memory Usage:** Efficient handling of a fixed number of objects

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7. Learning Outcomes

This project successfully demonstrates:

- **Inheritance** – Reuse of base structure across spot types
- **Polymorphism** – Different behaviors via a common interface
- **Encapsulation** – Controlled access to data
- **Dynamic Memory Management** – Runtime allocation and cleanup
- **Use of STL** – Efficient data handling
- **Structured Design** – Clean separation of responsibilities

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8. Error Handling & Reliability

The system includes safeguards for:

- Invalid user inputs
- Attempting to park in unavailable spots
- Incorrect spot IDs during exit
- Maintaining system stability during all operations

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9. Possible Improvements

Future enhancements could include:

- Persistent storage using file handling
- Smarter memory handling using modern techniques
- Faster search mechanisms
- Multi-user or concurrent operation support
- Graphical user interface instead of console

Sample Run & Output Explanation:

System Initialization and Main Menu

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 1
Parking lot initialized with 20 spots!
```

This screen displays the main menu of the Smart Parking Lot System, showing all available operations for the user. Upon selecting initialization, the system sets up the parking lot with a fixed capacity of 20 slots.

Standard Vehicle Parking Allocation

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 2
Enter parking hours: 2
Vehicle parked in S1 (Standard parking spot - $2/hour)
```

This screen shows the process of parking a standard vehicle by entering the required parking duration. The system assigns the first available standard slot (S1) and confirms the parking with the applicable hourly rate.

Disabled Parking with Discounted Rate

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 3
Enter parking hours: 4
Vehicle parked in D1 (Disabled parking spot - $0.50/hour (discounted))
```

This screen demonstrates parking a vehicle in a reserved disabled slot after entering the parking duration. The system allocates slot D1 and applies a reduced hourly rate, confirming discounted parking for eligible users.

VIP Parking with Premium Pricing

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 4
Enter parking hours: 6
Vehicle parked in V1 (VIP parking spot - $10 + $5/hour (premium))
```

This screen shows the allocation of a VIP parking slot after entering the parking duration. The system assigns slot V1 and applies premium pricing, including a base fee and higher hourly charges.

Parking Lot Status Overview

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 6

=== PARKING LOT STATUS ===
==================================================
Spot ID: S1, Type: Standard, Status: OCCUPIED
Spot ID: S2, Type: Standard, Status: AVAILABLE
Spot ID: S3, Type: Standard, Status: AVAILABLE
Spot ID: S4, Type: Standard, Status: AVAILABLE
Spot ID: S5, Type: Standard, Status: AVAILABLE
Spot ID: S6, Type: Standard, Status: AVAILABLE
Spot ID: S7, Type: Standard, Status: AVAILABLE
Spot ID: S8, Type: Standard, Status: AVAILABLE
Spot ID: S9, Type: Standard, Status: AVAILABLE
Spot ID: S10, Type: Standard, Status: AVAILABLE
Spot ID: S11, Type: Standard, Status: AVAILABLE
Spot ID: S12, Type: Standard, Status: AVAILABLE
Spot ID: D1, Type: Disabled, Status: OCCUPIED
Spot ID: D2, Type: Disabled, Status: AVAILABLE
Spot ID: D3, Type: Disabled, Status: AVAILABLE
Spot ID: D4, Type: Disabled, Status: AVAILABLE
Spot ID: V1, Type: VIP, Status: OCCUPIED
Spot ID: V2, Type: VIP, Status: AVAILABLE
Spot ID: V3, Type: VIP, Status: AVAILABLE
Spot ID: V4, Type: VIP, Status: AVAILABLE
==================================================
Total Spots: 20
Occupied Spots: 3
Occupancy: 15.0%
==================================================
```

This screen displays a detailed status of all parking slots, including their type and availability. It also provides a summary with total slots, occupied count, and overall occupancy percentage for monitoring usage.

Vehicle Exit and Fee Calculation

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 5
Enter Spot ID (e.g., V1, S5, D2): V1
Enter parking hours: 6
Vehicle exited from V1 (VIP)
Parking duration: 6 hours
Total fee: $40.00
--------------------------------------------------
```

This screen shows the process of exiting a parked vehicle by entering its slot ID and parking duration. The system calculates the total fee based on the parking type and duration, then frees the slot for future use.

System Exit and Termination

```
=== SMART PARKING LOT SYSTEM ===
1. Initialize Parking Lot
2. Park Vehicle (Standard)
3. Park Vehicle (Disabled)
4. Park Vehicle (VIP)
5. Exit Vehicle
6. Display Status
0. Exit
Enter choice: 0
Thank you for using Smart Parking Lot System!
```

This screen shows the user selecting the exit option to terminate the program. The system gracefully ends the session and displays a closing message to the user.